

The Kalman Filter from First Principles

1 · The model

Linear Gaussian state-space: $x_t = A x_{t-1} + w_t$, $z_t = H x_t + v_t$,
with $w \sim N(0, Q)$, $v \sim N(0, R)$. The filter maintains $p(x_t | z_{1:t}) = N(\hat{x}_t, P_t)$.

2 · Prediction step

$$\hat{x}_{t|t-1} = A \hat{x}_{t-1}, \quad P_{t|t-1} = A P_{t-1} A^T + Q$$

3 · Update (correction) step

$$\text{Innovation: } y_t = z_t - H \hat{x}_{t|t-1}$$

$$\text{Kalman gain: } K_t = P_{t|t-1} H^T (H P_{t|t-1} H^T + R)^{-1}$$

$$\text{Posterior: } \hat{x}_t = \hat{x}_{t|t-1} + K_t y_t$$

$$P_t = (I - K_t H) P_{t|t-1}$$

4 · Why it works

The Kalman gain is the MAP/MMSE estimator: it weighs model vs. measurement by their covariance - a theme shared with Approximate Message Passing, where beliefs play the role of $(\hat{x}, P)$ pairs. Hover over the interactive derivations on the web version for step-by-step algebra.

Full interactive version with LaTeX, diagrams and a runnable notebook:

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